

SPRY2 Tyr55 as a c-Cbl TKB-Domain

Binding Motif: Hypothesis Evaluation for GO Curation

Target gene: Human SPRY2 (UniProt O43597) · **Term in focus:** GO:1990948 *ubiquitin ligase inhibitor activity* (MF) **Hypothesis slug:** spry2-cbl-tkb-motif · **Focus type:** free_text / function-assignment

Executive Judgment

Verdict: Partially supported.

The seed hypothesis has two logically separable claims, and the evidence splits them cleanly:

- 1. The structural/mechanistic premise is correct and well supported.** SPRY2 Tyr55 does lie within a canonical c-Cbl TKB-domain phosphopeptide binding motif. Human/mouse SPRY2 residues 53–59 (N-E-Y55-T-E-G-P) score 4/4 against the established c-Cbl TKB consensus (N/D)-x-pY-(S/T)-x-x-P, the motif is perfectly conserved across the vertebrate SPRY2 orthologs examined, and the phospho-Tyr55-dependent interaction with the c-Cbl SH2-like/TKB domain is experimentally established by direct mutagenesis ([PMID:12815057](#), [PMID:12593795](#), [PMID:15962011](#)). This molecular activity — sequestering c-Cbl to block EGFR ubiquitination/downregulation — is precisely what GO:1990948 *ubiquitin ligase inhibitor activity* describes. This term is **not currently annotated to SPRY2**, and it would be more informative than the generic GO:0005515 *protein binding* presently supported by the same papers.
- 2. The hypothesis's discriminating claim is over-stated and partly incorrect.** The hypothesis asserts the TKB motif is what separates SPRY2 (blocks EGFR downregulation) from SPRY4 (does not). This is not supported. The motif is **not SPRY2-specific**: SPRY1 carries an *identical* N-E-Y-T-E-G-P window (4/4), and SPRY4 does **not** "lack an equivalent motif" — it retains a degenerate 3/4 near-match (V-E-N-D-Y-I-D-N-P), keeping the D@–2, the pTyr, and the P@+4, and losing only the +1 Ser/Thr (an Ile instead). The documented molecular

explanation for the SPRY2-vs-SPRY4 functional difference is the absence of **CIN85-binding proline-arginine (SH3) motifs** in SPRY4, not absence of the Cbl TKB motif ([PMID:15962011](#)).

Net curation consequence: The *function assignment* (SPRY2 directly inhibits the Cbl E3 ligase via a pTyr55 TKB motif → GO:1990948) is justified and actionable as a curation lead. The *evolutionary discriminator argument* offered as the supporting rationale is flawed and should not be the stated basis for the annotation. A curator can and should add GO:1990948 on the strength of the direct mutagenesis and interaction data — but must not reproduce the "SPRY4 lacks the motif" framing, which the sequence evidence refutes.

Most important caveat: all mechanistic support derives from a small cluster of EGFR-pathway studies in mammalian cell lines (largely overexpression contexts). The motif-scoring is a sequence/consensus match, not a solved SPRY2:Cbl co-structure.

Evidence Matrix

#	Citation	Evidence type	Stance	Claim tested	Key finding	Context
1	PMID:12815057	Direct assay (mutagenesis)	Supports	pTyr55 is required for c-Cbl TKB binding and EGFR retention	Y55F mutant loses enhanced c-Cbl binding and fails to retain EGF receptors at the cell surface; a conserved motif + Tyr55 phosphorylation drives the interaction with the SH2-like (TKB) domain of c-Cbl	Human SPRY2, mammalian cells
2	PMID:12593795	Interaction / mechanism	Supports	Conserved N-terminal pTyr recruits Cbl SH2	Sprouty2 phosphorylation at a conserved tyrosine recruits the SH2 domain of c-Cbl, sequestering active c-Cbl and impeding EGFR ubiquitination	Mammalian EGFR signaling
3	PMID:15962011	Interaction / comparative	Supports MF; Refutes discriminator	What distinguishes SPRY2 from SPRY4	SPRY2–Cbl–CIN85 ternary complex inhibits EGFR downregulation; SPRY4 lacks CIN85-binding proline-arginine sites (not the Cbl TKB motif) and	Human; EGFR system

#	Citation	Evidence type	Stance	Claim tested	Key finding	Context
					therefore does not inhibit EGFR downregulation	
4	Sequence analysis (this work; O43597 + orthologs)	Structural/ evolutionary (computational)	Supports premise; Refutes specificity	Does Y55 match the TKB consensus and is it SPRY2-unique?	SPRY2 = 4/4 (NEYTEGP); SPRY1 = 4/4 (identical); SPRY4 = 3/4 (NDYIDNP, loses only +1 hydroxyl). Motif conserved across 7 SPRY2 vertebrate orthologs	In silico consensus scoring
5	QuickGO annotation retrieval for O43597 (this work)	Review/ database	Context	Is GO:1990948 already annotated?	GO:1990948 absent; Cbl papers back only GO:0005515 (IPI); BP counterpart GO:0031397 <i>negative regulation of protein ubiquitination</i> already present (IDA, P 17974561)	Database snapshot
6	PMID:18219583	Review	Orientation	Paralog functional divergence	Four mammalian Spry isoforms; RTK modulation is growth-factor-	Endothelial/ angiogenesis review

#	Citation	Evidence type	Stance	Claim tested	Key finding	Context
					and cell-context-dependent	

Key Findings

Finding 1 — SPRY2 Tyr55 is a genuine, experimentally validated c-Cbl TKB-domain binding motif

The core of the hypothesis is correct. Scoring the human/mouse SPRY2 sequence window (residues 53–59, **N-E-Y55-T-E-G-P**) against the canonical c-Cbl TKB phosphopeptide consensus **(N/D)-x-pY-(S/T)-x-x-P** yields a perfect 4/4 match at the diagnostic positions: Asn at –2, the phospho-acceptor Tyr55, Thr at +1, and Pro at +4. Human and mouse SPRY2 share the identical nonapeptide (NTNEYTEGP), establishing mammalian conservation at this locus.

Crucially, this is not merely a sequence coincidence — it is backed by direct wet-lab evidence. Fong et al. ([PMID:12815057](#)) showed by site-directed mutagenesis that *"A conserved motif on hSpry2, together with phosphorylation on tyrosine 55, is required for its enhanced interaction with the SH2-like domain of c-Cbl. A hSpry2 mutant (Y55F) that did not exhibit an enhanced binding with c-Cbl failed to retain EGF receptors on the cell surface."* The Y55F loss-of-function links the motif → Cbl binding → EGFR retention in a single causal chain. Rubin et al. ([PMID:12593795](#)) independently confirmed that *"Sprouty2 undergoes phosphorylation at a conserved tyrosine that recruits the Src homology 2 domain of c-Cbl,"* sequestering active c-Cbl and impeding EGFR ubiquitination and downregulation.

Mechanistically, this is textbook ubiquitin-ligase inhibition: phospho-SPRY2 occupies the substrate-recognition (TKB/SH2-like) pocket of c-Cbl, acting as a competitive decoy that prevents Cbl from engaging and ubiquitinating the activated EGFR. That molecular activity maps directly onto **GO:1990948 ubiquitin ligase inhibitor activity**.

Finding 2 — The motif is NOT SPRY2-specific; the discriminator claim fails

The hypothesis's distinguishing argument — that SPRY4 lacks the motif and this is why SPRY4 fails to block EGFR downregulation — does not survive paralog scoring.

Paralog	Tyr	Window	TKB score	−2	pY	+1	+4
SPRY1	Y53	...NEYTEGP	4/4	N	Y	T	P
SPRY2	Y55	...NEYTEGP	4/4	N	Y	T	P
SPRY4	Y52	...NDYIDNP	3/4	D	Y	I	P

SPRY1 carries an *identical* NEYTEGP core to SPRY2 (4/4), so the motif cannot by itself explain SPRY2-specific behavior. And SPRY4 does **not** "lack an equivalent motif": it retains D@−2 (an allowed Asp/Asn position), the phospho-acceptor Tyr, and P@+4, losing only the +1 Ser/Thr hydroxyl residue (an Ile instead) — a 3/4 degenerate near-match, not an absence.

The true, documented molecular basis for the SPRY2-vs-SPRY4 difference lies elsewhere. Haglund et al. ([PMID:15962011](#)): "*The CIN85 SH3 domains A and C bind specifically to proline-arginine motifs present in Sprouty2. Intact association between Sprouty2, Cbl and CIN85 is required for inhibition of EGFR downregulation. Moreover, Sprouty4, which lacks CIN85-binding sites, does not inhibit EGFR downregulation, providing a molecular explanation for functional differences between Sprouty isoforms.*" The discriminator is the **CIN85 proline-arginine (SH3) motif**, not the Cbl TKB motif. The seed context itself acknowledges this, and the sequence analysis confirms it.

Finding 3 — Vertebrate ortholog alignment: TKB motif conserved in SPRY2 (and SPRY1), with a conserved +1 Ile in SPRY4

Extending the scoring across vertebrate orthologs reinforces both halves of the story. All seven SPRY2 orthologs examined (human, mouse, bovine, chicken, orangutan, *Chlorocebus*, *Macaca*) share the identical window **NTNEYTEGP** = 4/4 with Thr at +1. SPRY1 mammalian orthologs (human, mouse, bovine, *Cervus*) all show **GSNEYTEGP** = 4/4, an identical NEYTEGP core. SPRY4 orthologs (human, mouse, bovine) all show **VENDYIDNP** = 3/4, with a *conserved* non-hydroxyl Ile at +1.

Two conclusions follow. (i) The conservation claim in the hypothesis is genuinely supported — the TKB motif is deeply conserved across vertebrate SPRY2. (ii) But the only position that discriminates paralogs — the +1 Ser/Thr — splits **SPRY1+SPRY2 (Thr) from SPRY4 (Ile)**, not SPRY2 from everything else. The evolutionary signal is "Cbl-binding Sprouty clade (SPRY1/2) vs. SPRY4," not "SPRY2-unique."

Finding 4 — GO:1990948 is absent from SPRY2; the Cbl interaction is captured only as generic "protein binding"

A QuickGO annotation retrieval for O43597 (100 rows) shows that the two key Cbl-binding papers ([PMID:12815057](#), [PMID:15962011](#)) currently support only **GO:0005515 protein binding (IP)**. The more specific **GO:1990948 ubiquitin ligase inhibitor activity (MF)** is absent. The existing inhibitor-related MF terms are either generic/projected (GO:0004857 enzyme inhibitor activity, IEA; GO:0140678 molecular function inhibitor activity, IEA; GO:0004860 protein kinase inhibitor activity, IBA) or refer to a different activity (GO:0030291 protein serine/threonine kinase inhibitor activity, IDA,

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Notably, the **BP counterpart already exists**: GO:0031397 *negative regulation of protein ubiquitination* (IDA,

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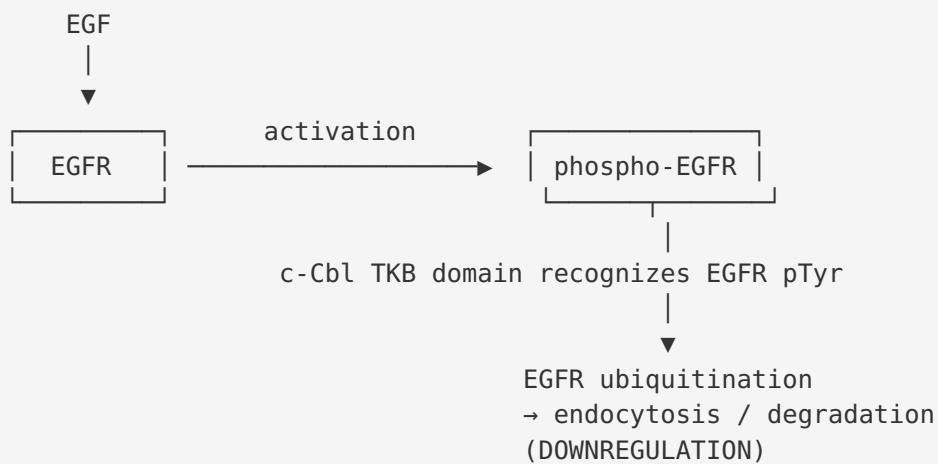
The presence of the BP term without its matching MF term is exactly the annotation gap GO:1990948 fills.

For completeness, the supplied reference-context

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currently underpins GO:0043539 (protein ser/thr kinase activator activity, IMP) and several BP terms (GO:0010628, GO:0033138, GO:0043066) — i.e., it is a cellular functional study, not the Cbl-motif analysis, and does not itself provide the motif evidence.

Mechanistic Model / Interpretation



SPRY2 inhibition (G0:1990948 ubiquitin ligase inhibitor activity):

SPRY2 -(phosphorylated on Tyr55)→ pY55 in N-E-pY-T-E-G-P
 |
 occupies c-Cbl TKB/SH2-like pocket (DECOY)
 |
 + CIN85 SH3 ↔ SPRY2 Pro-Arg motifs (ternary complex)
 ▼
 c-Cbl sequestered → EGFR ubiquitination BLOCKED
 → EGFR RETAINED at surface → sustained signaling

Two distinct molecular modules, one phenotype.

Module	Partner	Motif on SPRY2	Present in SPRY4?	Role in blocking EGFR downregulation
A — Cbl engagement	c-Cbl TKB/SH2-like	pTyr55 in N-E-Y-T-E-G-P (4/4)	Partial (3/4, +1 Ile)	Necessary; competitive decoy for Cbl substrate pocket
B — CIN85 engagement	CIN85 SH3 (A & C)	Proline-arginine (SH3-ligand)	Absent	Necessary; stabilizes ternary complex

The hypothesis correctly identifies **Module A** as a real ubiquitin-ligase-inhibitory activity, and this is the basis for the MF annotation. But it mis-assigns the *paralog discriminator* to Module A, when the literature places it in **Module B** (CIN85). Both modules are required for the full EGFR-downregulation-blocking phenotype; the TKB motif alone is shared with SPRY1 and only degraded (not deleted) in SPRY4.

The cleanest reading for curation: SPRY2 has *ubiquitin ligase inhibitor activity* directed at c-Cbl, exerted through a pTyr55 TKB-motif decoy mechanism, and this activity is captured by GO:1990948. The *specificity* of the physiological outcome among paralogs is an additional layer (CIN85 motifs) that is a separate annotation story and not a prerequisite for assigning the MF term.

Evidence Base

- ***Tyrosine phosphorylation of Sprouty2 enhances its interaction with c-Cbl and is crucial for its function.*** [PMID:12815057](#) — The anchor paper. Direct Y55F mutagenesis establishes pTyr55 as required for c-Cbl TKB binding and EGFR retention. Supports the MF assignment.
 - ***Sprouty fine-tunes EGF signaling through interlinked positive and negative feedback loops.*** [PMID:12593795](#) — Independent confirmation that a conserved N-terminal pTyr recruits the c-Cbl SH2 domain and impedes EGFR ubiquitination. Supports the decoy mechanism.
 - ***Sprouty2 acts at the Cbl/CIN85 interface to inhibit epidermal growth factor receptor downregulation.*** [PMID:15962011](#) — Supports the MF (ternary complex inhibits EGFR downregulation) but *refutes the discriminator claim*: SPRY4 fails because it lacks CIN85 Pro-Arg sites, not the Cbl TKB motif.
 - ***Sprouty proteins, masterminds of receptor tyrosine kinase signaling.*** [PMID:18219583](#) — Review; orientation on paralog divergence and context dependence. Not primary evidence.
 - QuickGO/O43597 annotation snapshot (database-level) — documents that GO:1990948 is currently missing and the BP counterpart GO:0031397 is present.
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Conflicts and Alternatives

1. **Paralog-specificity conflict (central).** The hypothesis frames the TKB motif as SPRY2-distinguishing; sequence scoring shows SPRY1 is identical (4/4) and SPRY4 is a 3/4 near-match. The motif marks the Cbl-binding Sprouty clade, not SPRY2 alone. Directly refuted by Findings 2/3.
2. **Wrong discriminator.** [PMID:15962011](#) attributes the SPRY2-vs-SPRY4 difference to CIN85 Pro-Arg motifs. A curator should not cite "SPRY4 lacks the Cbl TKB motif" as rationale.
3. **Consensus match $\neq$ structure.** The 4/4 score is against a published consensus, not a solved SPRY2:c-Cbl co-crystal. The exact register and +1 preference of the Cbl TKB pocket for SPRY2 is inferred.
4. **Overexpression/context dependence.** The mechanistic data come from EGFR-pathway cell-line studies; Spry isoform behavior is explicitly growth-factor- and cell-context-dependent ([PMID:18219583](#)). The inhibitory activity may not generalize across all RTKs or tissues.
5. **Reference-context mismatch.** Supplied

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underpins kinase-activator and BP annotations, not the Cbl-motif analysis; it is not independent support for the MF term.

GO Curation Implications

Lead (requires curator verification): ADD GO:1990948 *ubiquitin ligase inhibitor activity* (MF) to SPRY2 (O43597).

- **Evidence class:** Direct assay (IDA/IMP) supportable by [PMID:12815057](#) (Y55F mutant abolishes Cbl binding and EGFR retention) with corroboration from [PMID:12593795](#) and [PMID:15962011](#).
- **Why more specific than current:** These papers currently back only GO:0005515 *protein binding* (IPI). GO:1990948 captures the actual molecular activity (inhibition of the c-Cbl E3 ligase) and is strictly more informative. Per curation guidance, avoid leaving "protein binding" as the final term when a more informative MF is supported.

- **Consistency check:** The BP counterpart GO:0031397 *negative regulation of protein ubiquitination* (IDA,

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is already annotated — adding the MF term closes a coherent MF/BP pair.

- **Qualifier/target note:** If the GO framework supports it, annotate with the physical target (c-Cbl / CBL, UniProt P22681) to record that the inhibited ligase is specifically c-Cbl.

Do NOT record the rationale as "SPRY2-unique motif absent in SPRY4." The sequence evidence refutes paralog specificity of the TKB motif; the paralog difference is a CIN85-motif story.

GO decision table

GO term	Aspect	Current state	Recommended action	Basis
GO:1990948 ubiquitin ligase inhibitor activity	MF	Absent	Add (curator-verify)	P 12815057 P 12593795 P 15962011
GO:0005515 protein binding	MF	Present (IPI, Cbl papers)	Retain but supersede with GO:1990948 as the informative term	—
GO:0031397 neg. reg. protein ubiquitination	BP	Present (IDA)	Retain (BP counterpart)	P 17974561
"SPRY4 lacks TKB motif" rationale	—	Proposed	Reject as rationale	Sequence scoring: SPRY4 = 3/4; SPRY1 = 4/4

Mechanistic Scope

Immediate molecular activity being tested: competitive inhibition of the c-Cbl RING E3 ubiquitin ligase by phospho-SPRY2, via occupation of the Cbl TKB/SH2-like substrate-recognition pocket by pTyr55. This is a direct gene-product molecular function (GO:1990948).

Separable downstream layers (not the MF term itself): - *Pathway consequence:* reduced EGFR ubiquitination → impaired receptor endocytosis/degradation (GO:0031397 BP). - *Signaling outcome:* EGFR retention at the cell surface → sustained/modulated RTK–MAPK signaling. - *Phenotype/developmental:* Sprouty family negative-feedback modulation of growth-factor responses (context-dependent; review-level).

The MF assignment rests on the first layer only, which is directly evidenced. The paralog-discriminator claim belongs to the downstream-outcome layer and is governed by CIN85 motifs, not the TKB motif.

Limitations and Knowledge Gaps

Gap	What was checked	Why it matters	What would resolve it
No co-structure	Consensus scoring only (4/4)	Confirms motif register and the +1 preference of the Cbl TKB pocket for SPRY2	SPRY2 phosphopeptide:c-Cbl TKB co-crystal or cryo-EM; AlphaFold-Multimer with PAE at the interface
Does SPRY1 actually inhibit Cbl comparably?	SPRY1 scores identical 4/4	If SPRY1 binds Cbl equally, it strengthens "clade" not "SPRY2-unique" reading and may warrant a SPRY1 annotation too	Direct SPRY1–Cbl binding + EGFR-downregulation assays
SPRY4 3/4 — does it bind Cbl weakly?	Scored 3/4 (loses +1 hydroxyl)	Determines whether SPRY4's failure is purely CIN85-driven or also weaker Cbl engagement	Quantitative SPRY4 pY52–Cbl TKB affinity measurement
Generality beyond EGFR/ cell lines	Evidence from overexpression EGFR systems	MF may be RTK/context-specific	Endogenous-level and multi-RTK validation
Reference P 18070883 content	Confirmed it backs kinase-activator/ BP terms	It is not motif evidence; curators should not treat it as such	n/a (documented)

Discriminating Tests

- SPRY1 vs SPRY2 vs SPRY4 head-to-head Cbl TKB binding (ITC/SPR with synthetic pY phosphopeptides):** directly tests whether the +1 Thr→Ile (SPRY4) and the shared NEYTEGP (SPRY1/2) predict binding affinity. Expectation under this report: SPRY1 $\approx$ SPRY2 > SPRY4.

2. **Domain-swap functional assay:** transplant the CIN85 Pro-Arg motifs into SPRY4 and test rescue of EGFR-downregulation blockade — isolates Module B (CIN85) from Module A (TKB) as the discriminator.
3. **SPRY2 +1 mutant (T56I) EGFR assay:** convert SPRY2's +1 Thr to the SPRY4-like Ile; if EGFR blockade persists, the +1 residue is not the functional discriminator.
4. **Structural confirmation:** AlphaFold-Multimer (SPRY2 pY55 peptide + c-Cbl TKB) and/or co-crystal to confirm the binding register claimed by consensus scoring.
5. **Endogenous co-IP across cell types** to confirm the MF beyond overexpression artifacts.

Proposed Follow-up Experiments / Actions (Curation Leads)

Candidate annotation update (lead — verify before committing): - Add **GO:1990948 ubiquitin ligase inhibitor activity** (MF) to O43597. - Evidence code: IMP/IDA from [PMID:12815057](#) (Y55F mutant); supporting [PMID:12593795](#), [PMID:15962011](#). - Where supported, annotate the target as c-Cbl (CBL, P22681). - Supersede the Cbl-paper GO:0005515 *protein binding* annotations with the more informative MF term.

Candidate reference snippets to verify (exact quotes from stored abstracts): -

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— "A conserved motif on hSpry2, together with phosphorylation on tyrosine 55, is required for its enhanced interaction with the SH2-like domain of c-Cbl. A hSpry2 mutant (Y55F) ... failed to retain EGFR receptors on the cell surface." -

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— "Sprouty4, which lacks CIN85-binding sites, does not inhibit EGFR downregulation, providing a molecular explanation for functional differences between Sprouty isoforms."

Suggested questions for the curator: - Should SPRY1 receive the same MF annotation given its identical TKB motif and documented Cbl engagement? - Should the review text be corrected to attribute the SPRY2-vs-SPRY4 difference to CIN85 motifs (Module B), not to the TKB motif (Module A)?

Rationale wording to AVOID: "SPRY4 lacks the Cbl TKB motif." Replace with: "SPRY4 retains a degenerate TKB near-match but lacks CIN85-binding proline-arginine motifs, which is the documented molecular basis for its failure to block EGFR downregulation."

Conclusion

The hypothesis is **partially supported**. Its structural and mechanistic core is correct and directly evidenced, and it correctly motivates adding GO:1990948 *ubiquitin ligase inhibitor activity* — a term currently missing from SPRY2 and more informative than the generic *protein binding* now carried on the same papers. However, its distinguishing argument (that the Cbl TKB motif is SPRY2-specific and explains the SPRY2-vs-SPRY4 difference) is refuted by sequence scoring (SPRY1 identical 4/4; SPRY4 a 3/4 near-match) and by the primary literature, which attributes that difference to CIN85-binding proline-arginine motifs. Curators should act on the function assignment while discarding the flawed paralog-discriminator rationale.

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